

Experiment No.13 COUNTERS	
Course Code: CPE 013	Program: BSCpE
Course Title: Logic Circuits and Design	Date Performed: 14/4/2026
Section: CPE22S2	Date Submitted: 14/4/2026
Student Name: Bulaong, Ross Andrew D. Florencio, David Malabanan, Nadine Que, Leroy Josef Zapanta, Russell	Instructor: Engr. Noel Lusung
1. Objective(s):	
This activity aims to implement and demonstrate a sequential logic circuit – counter.	
2. Intended Learning Outcomes (ILOs):	
The students shall be able to: 2.1 Demonstrate and simulate the process of Synchronous and Asynchronous Counter in Logicy and TinkerCAD application. 2.2 Analyze and observe the process of each counter type.	
3. Discussion:	
A counter is a sequential logic circuit that goes through a prescribed sequence of states upon the application of input pulses. The prescribed sequence can be a binary sequence or any other sequence. A counter that goes through $2N$ (N is the number of flip-flops in the series) states is called a binary counter. The modulus of a counter is the number of different states it is allowed to have. Counter modulus is normally $2N$ unless controlled by a feedback circuit which limits the number of possible states (an example being the decimal counter). Counters are very widely used in almost all computers and other digital electronic systems. There are two major categories of counters: asynchronous counters and synchronous counters. - Asynchronous Counters Counters arranged so that the output of one flip-flop generates the clock input of the next higher stage are generally called asynchronous counters (or ripple counter). - Synchronous Counters Synchronous counters eliminate the cumulative flip-flop delay seen in ripple counter. Each flip-flop is clocked by the same clock signal. Each gate selectively controls when each more significant bit flip-flop is to change state (toggle) on the next clock transition.	
Source/Reference: http://ecelabs.njit.edu/ece394/lab6.php	

4. Resources:

- Personal Computer
 - Internet connection
- Simulation Tools:
- TinkerCAD
 - Logically

5. Procedure:

1. Construct and simulate Figure 13.1. Synchronous Counter in Logically.

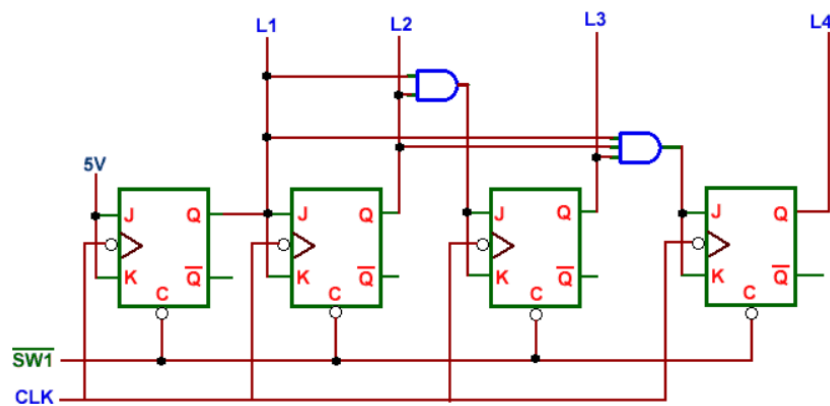


Figure 13.1. 4-Bit synchronous parallel counter

Image Source: <http://ecelabs.njit.edu/ece394/lab6.php>

2. Screenshot the simulation result and place it in data and results in section 6.
3. Repeat step 1 and 2 for Figure 13.2 but have the simulation in TinkerCAD.

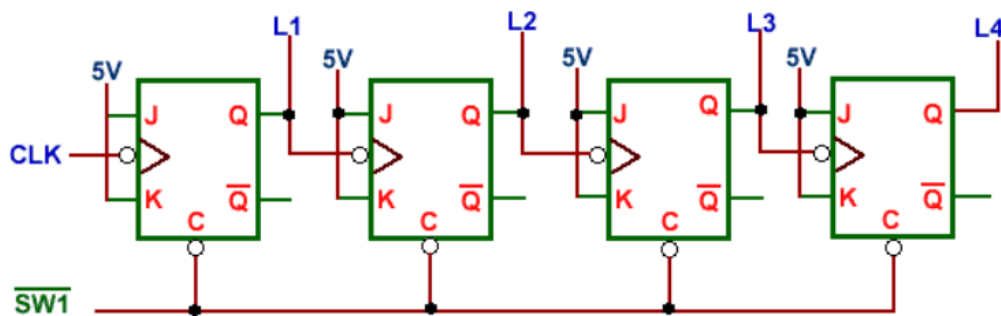


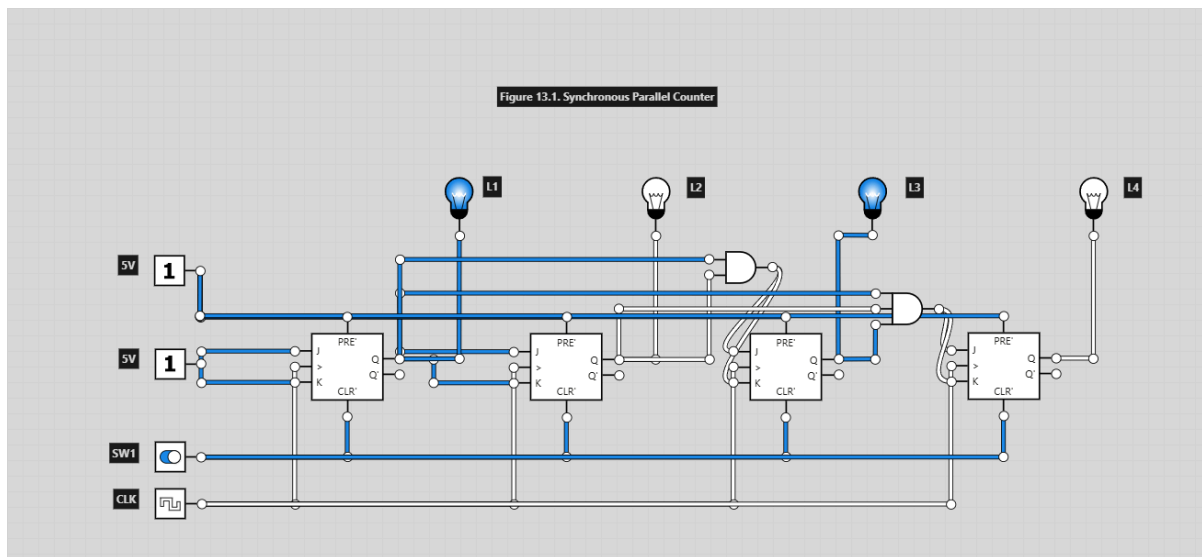
Figure 13.2. 4-Bit ripple counter

6. Data and Results:

Place here your simulation result for Figure 13.1 and Figure 13.2

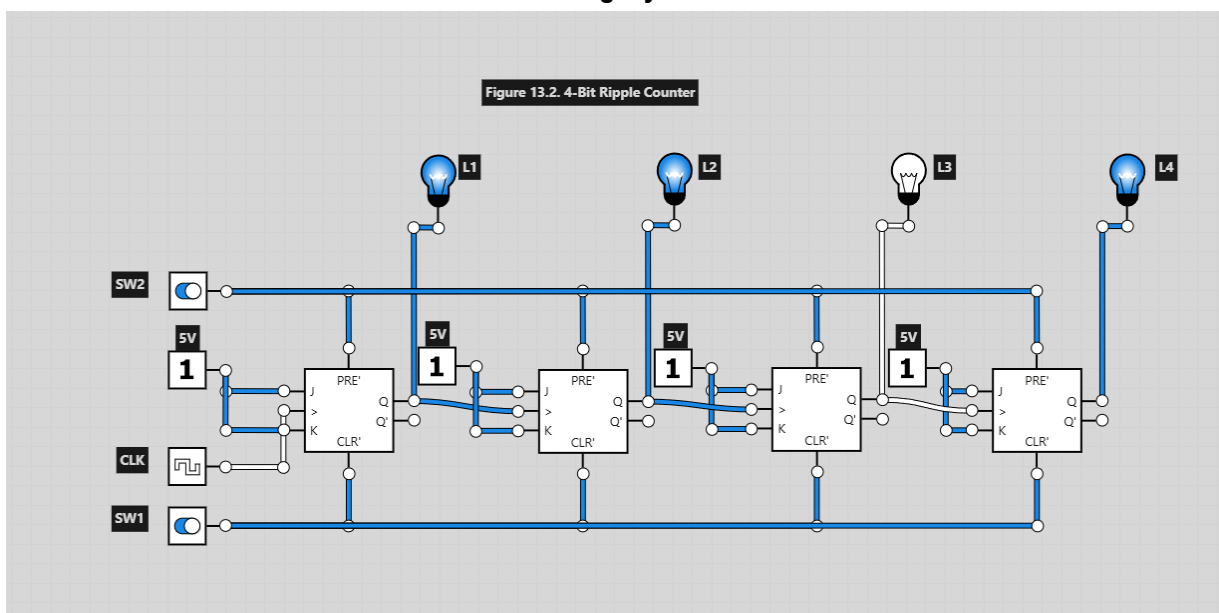
Circuit diagram for Figure 13.1

Logically

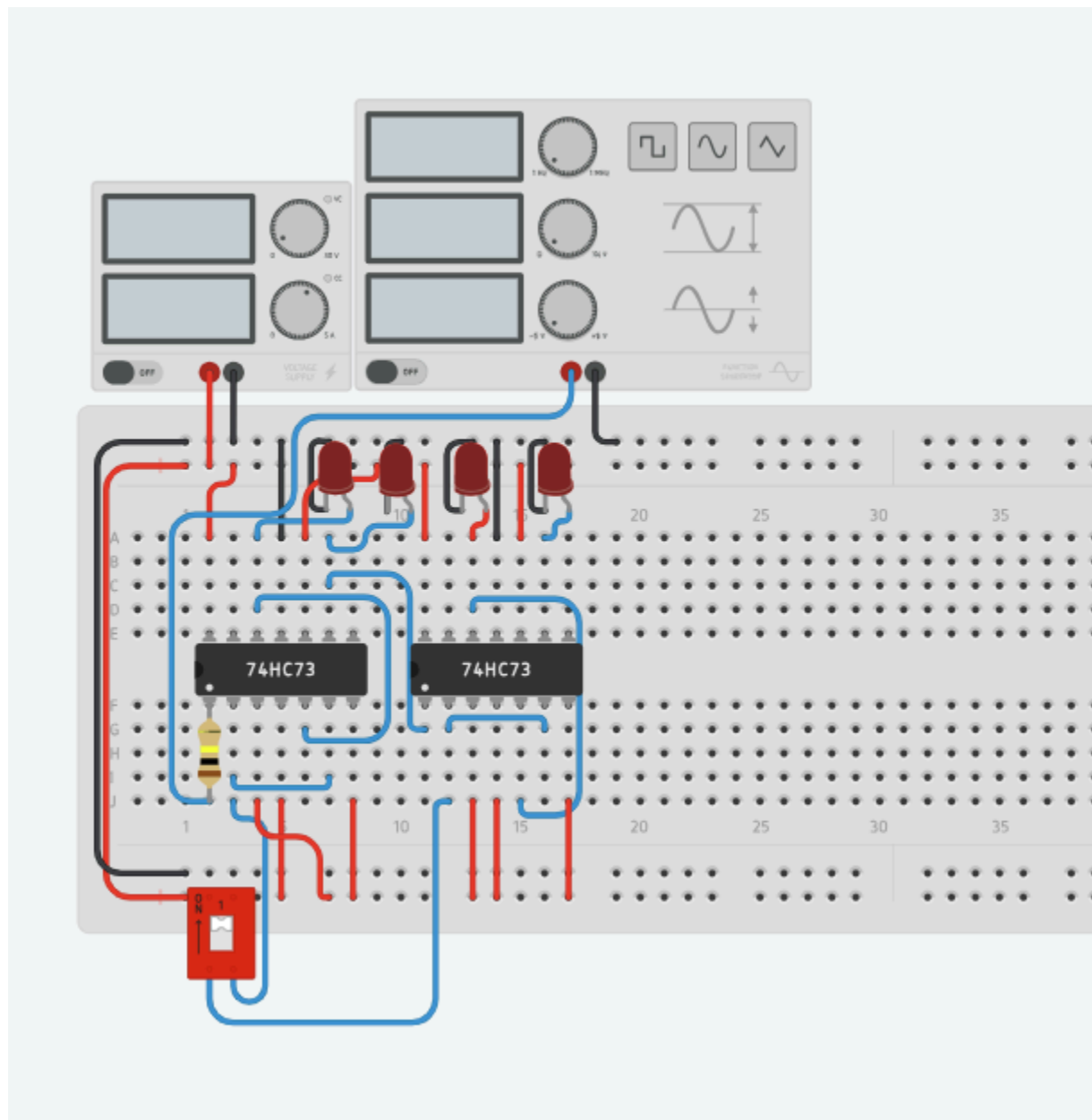


Circuit diagram for Figure 13.2

Logically



Tinkercad



7. Final Output:

Screenshot or print screen of Logically application different output simulations (5 points deductions for only one output in every required circuit). Also, upload a video of the simulated circuit naming or talking about the shown input combinations. Failure to comply to each instruction will result in 5 points deductions. Please include the recorded simulation video in this section and write at least one sentence definition or label for each figure or image.

<https://drive.google.com/drive/folders/1P9AcgSA2fr-zDivnxoUbH7NikA4zjdlb?usp=sharing>

<https://drive.google.com/drive/folders/16IOlCd6jSrVKOBim4ZSEzCNAw-yes2g6?usp=sharing>

8. Conclusion:

Write a one paragraph (at least 10 sentences) Conclusion for this activity.

This experiment demonstrated the design and simulation of both synchronous and asynchronous counters using digital tools. It helped in understanding how counters operate as sequential logic circuits with different timing behaviors. The synchronous counter proved to be more efficient due to its simultaneous switching and reduced delay. On the other hand, the asynchronous counter showed a ripple effect that affects speed and accuracy. Overall, the activity enhanced knowledge of counter circuits and their practical applications in digital systems.

9. Questions:

1. What is your observation for the simulation result in Figure 13.1 and 13.2? Expound your answer and evaluation.
 - In figure 13.1, the synchronous counter shows that all flip-flops change state the same time because they share a common clock signal, resulting in a faster and more stable output. In contrast, Figure 13.2 (ripple or asynchronous counter) shows a delay in the output transition since each flip flop depends on the previous one's output as its clock input. This delay creates a ripple effect, making the asynchronous counter slower and more prone to timing errors compared to the synchronous counter.
2. Cite different applications of counters in the industry.
 - Counters are widely used in digital clocks and watches to count seconds, minutes, and hours accurately. They are also applied in computer systems for memory addressing, instruction counting, and timing control processes. In industrial automation, counters are used in production lines to count items, control sequences, and monitor machine operations efficiently.

10. Assessment (Rubric for Laboratory Performance):

Laboratory Performance Rubric (2)					
Criteria	Ratings				Pts
Manipulative Skills	3 pts Proficient Student always demonstrate needed skills.	2 pts Acceptable Student occasionally demonstrate needed skills.	1 pts Beginner Student do not demonstrate needed skills.		3 pts
Experimental Set-up	3 pts Proficient Student are able to set-up the material with minimum supervision.	2 pts Acceptable Student are able to set-up the materials with supervision.	1 pts Beginner Student are unable to set-up the materials.		3 pts
Process Skills	3 pts Proficient Student always demonstrate targeted process skills.	2 pts Acceptable Student occasionally demonstrate targeted process skills.	1 pts Beginner Student do not demonstrate targeted process skills.		3 pts
Application of Technical Writing Skills	3 pts Proficient Correct spelling, grammar, and sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	2 pts Acceptable Some minor incorrect spelling, wrong grammar, and wrong sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.	1 pts Beginner Many incorrect spelling, wrong grammar, and wrong sentence construction when writing the Discussion, Observation, and Conclusion are applied in the laboratory report.		3 pts
Time Management / Conduct of Experiment	3 pts Proficient Student finish ahead of time with complete data and time to revise data.	2 pts Acceptable Student finish on time with incomplete data.	1 pts Beginner Student do not finish on time with incomplete data.		3 pts
Complete Requirements of Laboratory Report	3 pts Proficient Student submitted a complete requirements set by the teacher.	2 pts Acceptable Student submitted an almost complete Laboratory Report according to requirements set by the teacher.	1 pts Beginner Student submitted a lot of missing requirements in the Laboratory Report according to requirements set by the teacher.		3 pts
Follow the Correct Format of Laboratory Report	3 pts Proficient Student submitted a correct format of Laboratory Report according to requirements set by the teacher.	2 pts Acceptable Student submitted an almost correct format of Laboratory Report according to requirements set by the teacher.	1 pts Beginner Student did not follow at all the format of Laboratory Report according to requirements set by the teacher.		3 pts
Ability to do independent work	3 pts Proficient Student do not need to be supervised by the teacher.	2 pts Acceptable Student require occasional supervision by the teacher.	1 pts Beginner Student require supervision by the teacher.		3 pts
Total Points: 24					